Titanic Data Cleaning & Passenger Analysis

~Exploratory Data Analysis (EDA) Summary Report

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Objective

The objective of this project was to clean the Titanic dataset and analyze passenger survival trends using Exploratory Data Analysis (EDA). The project focused on handling missing values, removing duplicate records, identifying outliers, filtering passenger data, grouping records, calculating survival statistics, and visualizing important patterns that influenced passenger survival.

Dataset Information

Dataset Name: Titanic Dataset

Source: Kaggle

Dataset Link:
https://www.kaggle.com/datasets/yasserh/titanic-dataset

Dataset Description

The Titanic dataset contains information about passengers who travelled on the Titanic. It includes details such as passenger class, age, gender, fare, cabin information, embarkation point, and survival status. The dataset is widely used for learning data analysis and understanding the factors that affected passenger survival.

Data Exploration

The dataset was explored using functions such as head(), shape, info(), and describe(). These functions helped understand the structure of the dataset, identify data types, examine the number of records, and generate descriptive statistics before beginning the analysis.

Data Cleaning

The dataset was checked for missing values and duplicate records. Missing values in columns such as Age, Cabin, and Embarked were handled appropriately. Duplicate records were identified and removed to improve the quality and consistency of the dataset before analysis.

Statistical Analysis

Descriptive statistical measures were calculated for numerical columns such as Age and Fare. The analysis included mean, median, mode, variance, and standard deviation. These measures helped understand the distribution and variation of passenger-related data.

Outlier Detection

Outliers were detected using the Interquartile Range (IQR) method. Box plots were used to visualize extreme values in numerical columns such as Age and Fare. Identifying outliers helped understand unusual passenger records that could influence the analysis.

Filtering and Grouping

Passenger records were filtered based on age, gender, and passenger class. The groupby() function was used to analyze survival rates across different categories. This helped compare survival patterns among male and female passengers, different age groups, and passenger classes.

Survival Analysis

Survival statistics were calculated to understand the proportion of passengers who survived and those who did not. The analysis showed how factors such as passenger class, gender, and age influenced the chances of survival.

Data Visualization

Different visualizations were created to better understand passenger information and survival trends. These included Bar Chart, Line Chart, Histogram, Scatter Plot, Correlation Heatmap, and Box Plot. The visualizations made it easier to compare passenger groups and identify important relationships within the dataset.

Key Insights

The dataset was successfully cleaned before analysis. Missing values and duplicate records were handled appropriately. Female passengers had a higher survival rate than male passengers, and passengers travelling in First Class showed better survival rates compared to other classes. Visualizations clearly highlighted the factors that influenced passenger survival.

Conclusion

This project provided practical experience in data cleaning and Exploratory Data Analysis using the Titanic dataset. It covered data preprocessing, filtering, grouping, statistical analysis, outlier detection, survival analysis, and visualization. The project demonstrated how EDA can be used to understand real-world datasets and identify meaningful patterns that support data-driven conclusions.

Tools Used

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Google Colab

Project Outcome

Through this project, I gained hands-on experience in:

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Data Filtering
- Data Grouping
- Statistical Analysis
- Survival Analysis
- Outlier Detection

- Data Visualization
- Interpreting Analytical Results